

$S^n$ ,  ~~$S^1$~~ ,  $RP^n$ ,  $\langle \dots \rangle$   
 $(n \geq 1)$   
 $\downarrow$   
 $\pi_1 = 0$ , 2-dim mfd,  $L(p,q)$

$M \cup N$ ,  $\wedge \vee$ ,  $\langle S, S' \rangle / \langle e^{2\pi i p}, e^{2\pi i q} \rangle$   
 $\tau_0, \tau_1^2$ ,  $\mathbb{Z} \times \mathbb{Z}$ ,  $\langle a, b \mid aba = b \rangle$ ,  $ababab \dots$   
 $abab^t = e$ ,  $ab = ba \rightarrow$  group  $\mathbb{Z} \times \mathbb{Z}$ ,  $\langle a, b \mid ab = ba \rangle$ ,  $\mathbb{Z} \times \mathbb{Z}$   
 $S^1 \vee S^2$ ,  $T: \mathbb{Z} \times \mathbb{Z}$ ,  $T \hookrightarrow \mathbb{Z} \times \mathbb{Z}$ ,  $\mathbb{Z} \hookrightarrow \mathbb{Z} \times \mathbb{Z}$

Theorem:  $M: n$ -dim with  $(n \geq 2)$ , then  $\pi_1(M) \cong \pi_1(M \setminus \{x\})$ .  
 $\pi_1(RP^n) \cong \mathbb{Z}_2$  ( $n \geq 2$ ),  $\pi_1(S^n) = 0$  ( $n \geq 2$ ),  $\pi_1(R^3) = 0$ .

$\langle \dots \rangle$ ,  $\langle S, S' \rangle / \langle \dots \rangle$ ,  $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ ,  $S^2/p = S$ ,  $S^1: 0$ -cell  $\mathbb{Z}$ ,  $1$ -cell  $\mathbb{Z}$ ,  $S^2: 0$ -cell  $\mathbb{Z}$ ,  $2$ -cell  $\mathbb{Z}$ .

Example:  $RP^2: 0$ -cell:  $\mathbb{Z}$ ,  $1$ -cell:  $\mathbb{Z}$ ,  $2$ -cell:  $\mathbb{Z}$ ,  $\langle a \mid a^2 \rangle$ ,  $RP: 0$ -cell  $\mathbb{Z}$ ,  $1$ -cell  $\mathbb{Z}$ ,  $2$ -cell  $\mathbb{Z}$ ,  $\pi_1(RP^2) = \pi_1(RP) = \mathbb{Z}_2$ .

$RP^2: \mathbb{Z}_2$ ,  $\langle a \mid a^2 \rangle$ ,  $a^2 = 1$ ,  $\pi_1(RP^2) = \pi_1(RP) = \mathbb{Z}_2$ ,  $\mathbb{Z} = \mathbb{Z} + \mathbb{Z} + \mathbb{Z}$ ,  $\cup \mathbb{Z} \times \mathbb{Z}$ ,  $\cup (0,0,2) \rightarrow \mathbb{Z} \times \mathbb{Z}$ .

$\mathbb{R}/\mathbb{Z}$ ,  $\mathbb{Z} = \mathbb{Z} + \mathbb{Z}$ ,  $\mathbb{Z} = \mathbb{Z} + \mathbb{Z}$ ,  $\mathbb{Z} = \mathbb{Z} + \mathbb{Z}$ ,  $\mathbb{Z} = \mathbb{Z} + \mathbb{Z}$ .

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